Feasibility and Causal Information Audit for the POLARIS 2024 Nuolja Irradiance Benchmark

1. Baseline Context and the 28\% n\text{MAE} Information Bottleneck

The operational forecasting benchmark for the POLARIS project is currently held by the Stage 7.5 Terrain/Horizon champion model. Evaluated across the locked 173 ODD held-out test origins comprising 2,883 daytime forecast hours at SITES Nuolja Middle Station (68.3621^\circ\text{N}, 18.7388^\circ\text{E}, altitude \sim 675\text{ m a.s.l.}), Stage 7.5 achieved a daytime Normalized Mean Absolute Error (n\text{MAE}) of 28.3983\%, a daytime Mean Absolute Error (\text{MAE}) of 36.613571\text{ W/m}^2, a Root Mean Squared Error (\text{RMSE}) of 65.347064\text{ W/m}^2, and a daytime bias of +1.307778\text{ W/m}^2. Compared to the preceding Stage 6.6 LWP/IWP champion (28.7026\% \text{ nMAE}, 37.005888\text{ W/m}^2 \text{ MAE}), Stage 7.5 produced an observed held-out reduction of -0.3043 percentage points in n\text{MAE} and -0.392318\text{ W/m}^2 in daytime \text{MAE}. However, a 10,000-resample paired origin-level bootstrap yielded a mean \Delta\text{MAE} of -0.397063\text{ W/m}^2 with a 95\% confidence interval of [-0.791521, +0.001941]\text{ W/m}^2. Because the upper bound of this confidence interval crosses zero, the observed improvement cannot be classified as statistically significant under the specified bootstrap procedure, and Stage 7.5 remains frozen as an operational baseline.

Benchmark Stage	Model Family	Key Input Additions	ODD Daytime MAE (\text{W/m}^2)	ODD Daytime n\text{MAE} (\%)	Statistical Status vs. Previous Champion
Stage 5.2	GBDT Residual Corrector	Base 28 surface/solar/NWP features	39.084821	30.3151%	Retired Operational Baseline
Stage 6.6	GBDT Residual Corrector	+ LWP, IWP, LWP \times Cloud interaction	37.005888	28.7026%	Statistically Significant ($p < 0.0001$)
Stage 7.5	GBDT Residual Corrector	+ DEM 30m 360° Terrain Horizon Features	36.613571	28.3983%	Observed Improvement (CI crosses zero)

An exhaustive analysis of the experimental lineage demonstrates that POLARIS has encountered a structural information barrier at approximately 28\% \text{ nMAE}. To date, extensive feature engineering and data extraction attempts have evaluated and rejected multiple candidate feature spaces:

- **MEPS Ensemble Spread (Stage 6.1):** Demonstrated strong physical correlation with

error magnitude (\rho = +0.7670), but introducing summary statistics directly into gradient-boosted decision trees (GBDT) induced mild out-of-sample overfitting (+0.0457\text{ W/m}^2 \text{ CV} degradation).
- **Thermal-IR Satellite Brightness Temperatures (Stage 6.2):** Point-based 11\ \mu\text{m} and 12\ \mu\text{m} brightness temperatures failed forward-chaining fold stability tests (+0.0290\text{ W/m}^2 \text{ CV} degradation) due to coarse sensor resolution and orbital temporal gaps.
- **MEPS Forecast Revisions (Stage 6.4):** Sequential forecast update trajectories (f_1 \dots f_4) contained measurable predictive signal but exhibited severe fold instability (+0.2857\text{ W/m}^2 degradation in Fold 1).
- **MEPS Native Surface Radiation (Stage 7.2):** Instantaneous and hourly accumulated shortwave outputs from the MEPS RRTMG scheme provided redundant information relative to existing predictors, yielding +0.2083\text{ W/m}^2 to +0.4412\text{ W/m}^2 \text{ CV} degradations.
- **Clear-Sky Index (k_t) Target Reformulation (Stage 7.3):** Normalizing target irradiance by clear-sky potential created extreme heavy-tailed ratio instabilities during low-sun transitions (\text{GHI}_{\text{clear}} \to 0), causing severe degradation (+1.7020\text{ W/m}^2 in Fold 3).
- **Vertical Cloud Sub-Layer Structuring (Stage 7.4):** Partitioning liquid and ice mass into low, mid, and high pressure-level tiers failed all five forward-chaining acceptance gates (+0.0063\text{ W/m}^2 \text{ CV} degradation).
- **Spatial NWP Neighborhood Probes (Stage 5.4):** Offset grid probes and surface wind vectors (U_{10}, V_{10}) produced out-of-sample degradation (+0.5804\text{ W/m}^2 on ODD test origins).

The fundamental scientific cause of this 28\% \text{ nMAE} barrier is an information deficit regarding dynamic cloud optical depth (\tau) and 3D atmospheric cloud advection. Point observations at SITES Nuolja, static topographic horizon masks, and 1D column-integrated hydrometeor totals provide zero spatial visibility into upwind cloud fields moving across the Arctic terrain. Breaking the 28\% barrier requires identifying new, causally eligible data sources that explicitly capture upstream cloud mass trajectories, cloud phase microphysics, or high-resolution optical extinction prior to forecast origin time t_0.

2. Exhaustive Variable-by-Variable Data Source Audit

To evaluate potential new causal information sources for the 2024 Nuolja forecasting benchmark, candidate datasets were audited against strict archival, spatial, temporal, and causality criteria. Each candidate is analyzed below across seventeen standardized technical properties (A through Q) to verify actual availability rather than theoretical inclusion in CF standard registries.

2.1 MEPS / MET Norway Model Archives

Candidate 1: MEPS Native Model-Level 3D Hydrometeor Mass Fractions

- **A. Exact Dataset/Product Name:** MET Norway MEPS Deterministic 3D Vertical Collection (meps_det_vc_*.ncml).
- **B. Official Provider:** MET Norway (Norwegian Meteorological Institute) via THREDDS

Service.

- **C. Exact Variable Name:** mass_fraction_of_cloud_liquid_water_in_air (q_l), mass_fraction_of_cloud_ice_water_in_air (q_i) on hybrid model levels 1 to 65.
- **D. Spatial Resolution:** 2.5 km horizontal grid; 65 terrain-following hybrid vertical levels (\eta-coordinates) concentrated in the boundary layer.
- **E. Temporal Resolution:** 1-hour forecast time steps out to +67 hours.
- **F. Coverage at Nuolja:** 100\% geographic coverage at target grid point y=760, x=486 (68.3557^\circ\text{N}, 18.7637^\circ\text{E}, distance 1.31 km).
- **G. Historical 2024 Availability:** Fully archived for all 347 benchmark origins in 2024.
- **H. Archive Access Method:** NetCDF Subset Service (NCSS) / OPENDAP via thredds.met.no.
- **I. Forecast vs. Analysis/Observation:** Numerical Weather Prediction (NWP) forecast trajectory.
- **J. Forecast Initialization Time:** Issued at 00:00, 06:00, 12:00, 18:00 UTC. Causal origin selection uses D-1\text{ 18:00 UTC} (t_{\text{ref}} = t_0 - 6\text{h}).
- **K. Causally Available at t_0:** Yes; t_{\text{ref}} < t_0 is strictly satisfied across all origins.
- **L. Dissemination Latency:** Documented at \sim 2.5 to 3.0 hours post-initialization. The 18:00 UTC run is fully disseminated by 21:00 UTC, well before t_0 = \text{00:00 UTC}.
- **M. Already Tested in POLARIS:** No. Stage 6.6 tested vertical integrals on standard pressure levels (1000 hPa to 300 hPa). Model-level \eta-coordinate fields in the lowest 1500 m above terrain remain untested.
- **N. Expected Information Gain:** High. Nuolja station altitude (\sim 675\text{ m a.s.l.}, surface pressure \sim 930\text{ hPa}) falls between standard pressure levels. Native hybrid levels provide resolved boundary-layer cloud mass directly adjacent to the mountain topography.
- **O. Storage/Download Burden:** High. Extracting 65 vertical levels across a spatial grid requires \sim 450\text{ MB} per forecast origin.
- **P. Technical Extraction Difficulty:** High. Requires evaluating hybrid vertical coordinates p(k) = a(k) + b(k) \cdot p_{\text{sfc}} dynamically using time-varying surface pressure p_{\text{sfc}}.
- **Q. Leakage Risk:** Zero, provided t_{\text{ref}} < t_0 alignment is maintained.

Candidate 2: MEPS Cloud Base and Cloud Top Altitude Diagnostics

- **A. Exact Dataset/Product Name:** MET Norway MEPS Deterministic Surface Collection (meps_det_sfc_*.ncml).
- **B. Official Provider:** MET Norway.
- **C. Exact Variable Name:** cloud_base_altitude (m a.s.l.), cloud_top_altitude (m a.s.l.).
- **D. Spatial Resolution:** 2.5 km horizontal grid.
- **E. Temporal Resolution:** 1-hour forecast steps.
- **F. Coverage at Nuolja:** 100\% complete at target grid point.
- **G. Historical 2024 Availability:** Fully archived in 2024 surface netCDF files.
- **H. Archive Access Method:** OPENDAP / NCSS point extraction via thredds.met.no.
- **I. Forecast vs. Analysis/Observation:** NWP forecast diagnostic.
- **J. Forecast Initialization Time:** D-1\text{ 18:00 UTC} (t_{\text{ref}} < t_0).
- **K. Causally Available at t_0:** Yes.
- **L. Dissemination Latency:** Disseminated prior to t_0.
- **M. Already Tested in POLARIS:** No.

- **N. Expected Information Gain:** Medium. Explicit geometric cloud thickness (\Delta h = h_{\text{top}} - h_{\text{base}}) and sub-cloud layer depth dictate diffuse light scattering and ground-level clearance.
- **O. Storage/Download Burden:** Very Low (< 2\text{ MB} for 347 origins).
- **P. Technical Extraction Difficulty:** Low. Standard 2D surface field extraction.
- **Q. Leakage Risk:** Zero.

Candidate 3: MEPS Mid-Troposphere Steering Wind Fields

- **A. Exact Dataset/Product Name:** MET Norway MEPS Deterministic Pressure-Level Collection (meps_det_pl_*.ncml).
- **B. Official Provider:** MET Norway.
- **C. Exact Variable Name:** x_wind (U), y_wind (V) at 700\text{ hPa} and 850\text{ hPa} pressure levels.
- **D. Spatial Resolution:** 2.5 km horizontal grid.
- **E. Temporal Resolution:** 1-hour forecast steps.
- **F. Coverage at Nuolja:** 100\% coverage at target grid point.
- **G. Historical 2024 Availability:** Fully archived in 2024 pressure-level files.
- **H. Archive Access Method:** OPENDAP / NCSS point extraction via thredds.met.no.
- **I. Forecast vs. Analysis/Observation:** NWP forecast trajectory.
- **J. Forecast Initialization Time:** D-1\text{ 18:00 UTC} (t_{\text{ref}} < t_0).
- **K. Causally Available at t_0:** Yes.
- **L. Dissemination Latency:** Disseminated prior to t_0.
- **M. Already Tested in POLARIS:** Partially. Stage 5.4 tested 10\text{ m} surface winds (U_{10}, V_{10}), which were severely distorted by local mountain friction. Free-atmosphere steering winds at 700\text{ hPa} (\sim 3000\text{ m a.s.l.}) have not been tested.
- **N. Expected Information Gain:** High. Steering winds determine the advection vector of upwind cloud fields relative to Mt. Nuolja's topographic ridge.
- **O. Storage/Download Burden:** Low (\sim 15\text{ MB} for 347 origins).
- **P. Technical Extraction Difficulty:** Low. Standard 3D pressure-level slicing.
- **Q. Leakage Risk:** Zero.

2.2 MET Nordic Analysis and Forecast System

Candidate 4: MET Nordic Gridded Surface Analysis

- **A. Exact Dataset/Product Name:** MET Nordic Combined Reanalysis/Analysis Grid (met_nordic_analysis_*.nc).
- **B. Official Provider:** MET Norway, Swedish Meteorological and Hydrological Institute (SMHI), Finnish Meteorological Institute (FMI).
- **C. Exact Variable Name:** surface_downwelling_shortwave_flux_in_air, air_temperature_2m, relative_humidity_2m.
- **D. Spatial Resolution:** 1.0 km spatial grid covering Fennoscandia.
- **E. Temporal Resolution:** 1-hour analysis steps.
- **F. Coverage at Nuolja:** 100\% geographic coverage.
- **G. Historical 2024 Availability:** Archived in SMHI Open Data and MET Norway THREDDS archives.

- **H. Archive Access Method:** HTTPS / REST API retrieval.
- **I. Forecast vs. Analysis/Observation:** Surface analysis / spatial reanalysis incorporating station observations, radar, and satellite nudging.
- **J. Forecast Initialization Time:** Analysis timestamp t_{\text{an}}.
- **K. Causally Available at t_0: Partially / Conditional.** Analysis values for t \le t_0 are causally available as historical ground context. Analysis values for t > t_0 (H1 \dots H24) represent future observations and are **strictly causally ineligible**.
- **L. Dissemination Latency:** Documented at +45 to +60 minutes post-observation hour.
- **M. Already Tested in POLARIS:** No.
- **N. Expected Information Gain:** Low for future forecast steps (unusable); Moderate as pre-forecast context (t \le t_0) due to 1.0 km resolution terrain matching.
- **O. Storage/Download Burden:** Moderate (\sim 15\text{ GB} for regional spatial grid time-series).
- **P. Technical Extraction Difficulty:** Medium. Requires reprojecting 1 km Nordic grid to Nuolja target coordinates.
- **Q. Leakage Risk: EXTREMELY HIGH.** Using MET Nordic analysis fields at t > t_0 introduces direct target leakage by incorporating observed station data into input features.

2.3 EUMETSAT Polar Satellite Products and NWC SAF/PPS

Candidate 5: NWC SAF/PPS Polar Satellite Cloud Optical & Microphysical Properties

- **A. Exact Dataset/Product Name:** EUMETSAT NWC SAF PPS Cloud Physical Properties (CPP) v2021/v2025 (MetOp-B/C, NOAA-19/20/21, Sentinel-3A/B).
- **B. Official Provider:** EUMETSAT / MET Norway Polar Satellite Archive.
- **C. Exact Variable Name:** cpp_cot (Cloud Optical Thickness), cpp_cwp (Cloud Water Path), cpp_phase (Cloud Phase).
- **D. Spatial Resolution:** \sim 1.0\text{ km} pixel resolution at nadir.
- **E. Temporal Resolution:** Irregular overpass times; \sim 10 to 14 polar overpasses per day over Nuolja (68.36^\circ\text{N}).
- **F. Coverage at Nuolja:** Highly intermittent. Overpasses occur at discrete times with variable orbit geometry.
- **G. Historical 2024 Availability:** Archived in MET Norway polar satellite directory.
- **H. Archive Access Method:** File-level NetCDF processing via MET Norway THREDDS polar-swath archives.
- **I. Forecast vs. Analysis/Observation:** Observational satellite retrieval.
- **J. Forecast Initialization Time:** Observation acquisition time t_{\text{sat}}.
- **K. Causally Available at t_0:** Conditional. Usable only if overpass acquisition time satisfies t_{\text{sat}} \le t_0 - \Delta t_{\text{proc}}.
- **L. Dissemination Latency:** Documented at 30 to 75 minutes for Level-2 polar swath processing.
- **M. Already Tested in POLARIS:** No. Stage 6.2 evaluated raw thermal-IR brightness temperatures (BT11, BT12) from AVHRR swaths; derived physical optical thickness (cpp_cot) was not tested.
- **N. Expected Information Gain:** High at origin time t_0, but degraded by orbital gaps across H1 \dots H24 lead times.

- **O. Storage/Download Burden:** High (\sim 120\text{ GB} regional swath archive).
- **P. Technical Extraction Difficulty:** Extremely High. Requires spatial point-in-polygon matching, orbital swath reprojection, and handling missing overpass origins.
- **Q. Leakage Risk:** Moderate. Requires strict verification that overpass time t_{\text{sat}} < t_0.

CRITICAL ARCHIVAL WARNING / LESSON LEARNED: A forensic audit of the Stage 6.2 extraction revealed that out of 347 benchmark origins, the MET Norway polar satellite archive yielded valid causal overpasses for only 334 origins. Furthermore, the extraction was overwhelmingly dominated by legacy AVHRR sensors, yielding **only 2 valid VIIRS overpasses** across the entire 2024 dataset. Promising satellite optical depth products frequently suffer from severe archival gaps in operational repositories.

Candidate 6: EUMETSAT NWC SAF High-Resolution Winds / Cloud Motion Vectors (HRW/CMV)

- **A. Exact Dataset/Product Name:** NWC SAF PPS High Resolution Winds (HRW) Product.
- **B. Official Provider:** EUMETSAT / NWC SAF.
- **C. Exact Variable Name:** amv_speed (Atmospheric Motion Vector Speed), amv_dir (Direction), amv_pressure (Cloud Height Pressure).
- **D. Spatial Resolution:** Discrete irregular vector point-cloud derived from consecutive satellite overpasses.
- **E. Temporal Resolution:** Overpass pair intervals (30 to 100 minutes).
- **F. Coverage at Nuolja:** Sparse, spatially discontinuous vector coverage over Scandinavia.
- **G. Historical 2024 Availability:** Partial archive in EUMETSAT Data Centre.
- **H. Archive Access Method:** EUMETSAT Data Store API / NWC SAF file retrieval.
- **I. Forecast vs. Analysis/Observation:** Observational derived vector product.
- **J. Forecast Initialization Time:** Observation pair time t_{\text{obs2}} \le t_0.
- **K. Causally Available at t_0:** Yes, for overpasses completed prior to t_0.
- **L. Dissemination Latency:** +45 to +90 minutes post-overpass.
- **M. Already Tested in POLARIS:** No.
- **N. Expected Information Gain:** High theoretical value for computing Lagrangian cloud advection toward Nuolja.
- **O. Storage/Download Burden:** Moderate (\sim 25\text{ GB}).
- **P. Technical Extraction Difficulty:** Extremely High. Requires dynamic Lagrangian trajectory modeling from irregular vector points to project upwind cloud arrival times at Nuolja.
- **Q. Leakage Risk:** Low, provided overpass timestamps are validated.

2.4 CAMS Global Atmospheric Composition and Radiation Services

Candidate 7: CAMS Global Atmospheric Composition Forecasts - Aerosol Optical Depth

- **A. Exact Dataset/Product Name:** CAMS Global Atmospheric Composition Forecasts (cams_global_atmospheric_composition_forecasts).

- **B. Official Provider:** European Centre for Medium-Range Weather Forecasts (ECMWF) / Copernicus Atmosphere Monitoring Service (CAMS).
- **C. Exact Variable Name:** aod550 (Total Aerosol Optical Depth at 550 nm), duaod550 (Dust AOD), omaod550 (Organic Matter AOD), suaod550 (Sulfate AOD).
- **D. Spatial Resolution:** 0.4^\circ \times 0.4^\circ (\sim 40\text{ km} spatial grid).
- **E. Temporal Resolution:** 3-hour forecast steps out to +120 hours.
- **F. Coverage at Nuolja:** 100\% global grid coverage (68.4^\circ\text{N}, 18.8^\circ\text{E}).
- **G. Historical 2024 Availability:** Fully archived in the Copernicus Atmosphere Data Store (ADS).
- **H. Archive Access Method:** Python cdsapi / ADS REST API retrieval.
- **I. Forecast vs. Analysis/Observation:** Global atmospheric chemistry NWP forecast.
- **J. Forecast Initialization Time:** Issued twice daily at 00:00 UTC and 12:00 UTC. Causal origin selection requires using the D-1\text{ 12:00 UTC} run (t_{\text{ref}} = t_0 - 12\text{h}) or D0\text{ 00:00 UTC} run depending on dissemination rules.
- **K. Causally Available at t_0:** Yes, when using D-1\text{ 12:00 UTC} initialization.
- **L. Dissemination Latency: CRITICAL DISSEMINATION LATENCY FACTOR.** The CAMS 00:00 UTC forecast run undergoes complex chemical transport modeling and is disseminated at approximately 10:00 UTC. Therefore, at t_0 = \text{00:00 UTC}, the D0\text{ 00:00 UTC} CAMS run is **NOT available**. Operational causal eligibility requires using the D-1\text{ 12:00 UTC} forecast run.
- **M. Already Tested in POLARIS:** No.
- **N. Expected Information Gain:** Low to Moderate. Arctic baseline aerosol optical depth at Nuolja is pristine (\text{AOD}_{550} \sim 0.03 - 0.08), except during wildfire smoke transport events (e.g., Siberian/Canadian smoke plumes). Aerosol variations account for 1 - 5\text{ W/m}^2 clear-sky modulation, which is secondary to cloud extinction errors (100 - 400\text{ W/m}^2).
- **O. Storage/Download Burden:** Very Low (< 10\text{ MB} for point time-series across 347 origins).
- **P. Technical Extraction Difficulty:** Low. Standard ADS API extraction.
- **Q. Leakage Risk:** Moderate if CAMS 00:00 UTC run is incorrectly assumed available at 00:00 UTC.

Candidate 8: CAMS Solar Radiation Service (CAMS-RAD / McClear)

- **A. Exact Dataset/Product Name:** CAMS Radiation Service (McClear Clear-Sky Model / CAMS All-Sky).
- **B. Official Provider:** ECMWF / CAMS / Transvalor.
- **C. Exact Variable Name:** ghi_clear (McClear Clear-Sky GHI), bhi_clear (McClear Clear-Sky Beam Irradiance), dhi_clear (McClear Clear-Sky Diffuse Irradiance).
- **D. Spatial Resolution:** Interpolated point time-series at exact station coordinates (68.3621^\circ\text{N}, 18.7388^\circ\text{E}).
- **E. Temporal Resolution:** 1-minute to 1-hour temporal resolution.
- **F. Coverage at Nuolja:** 100\% station coverage.
- **G. Historical 2024 Availability:** Fully archived for 2024 via CAMS Radiation API.
- **H. Archive Access Method:** HTTPS REST API retrieval.
- **I. Forecast vs. Analysis/Observation: Analysis-Driven Observation Model.** CAMS-RAD McClear computes clear-sky radiation by ingesting CAMS reanalysis/analysis aerosol, water vapor, and ozone fields.

- **J. Forecast Initialization Time:** Non-forecast product; uses post-processed atmospheric composition analysis.
- **K. Causally Available at t_0: NO for future forecast hours (H1 \dots H24).** Because CAMS-RAD queries ingest atmospheric composition analysis fields computed post-facto, querying CAMS-RAD for future hours at t > t_0 constitutes **future observation leakage**. Only forecasted clear-sky models initialized at t_{\text{ref}} < t_0 are causally valid.
- **L. Dissemination Latency:** Reanalysis mode is delayed by 1 to 2 days.
- **M. Already Tested in POLARIS:** No. POLARIS utilizes the Solis and Ineichen clear-sky models computed locally from astronomical positioning.
- **N. Expected Information Gain:** Zero for operational forecasting due to causal invalidity for future lead times.
- **O. Storage/Download Burden:** Very Low (< 5\text{ MB}).
- **P. Technical Extraction Difficulty:** Low.
- **Q. Leakage Risk: HIGH.** Using historical CAMS-RAD all-sky or clear-sky outputs for H1 \dots H24 introduces severe reanalysis data leakage.

2.5 External Reanalysis and High-Resolution Radiation Products

Candidate 9: ERA5-Land Climate Reanalysis Shortwave Fluxes

- **A. Exact Dataset/Product Name:** ERA5-Land Hourly Data from 1950 to Present (reanalysis-era5-land).
- **B. Official Provider:** ECMWF / Copernicus Climate Change Service (C3S).
- **C. Exact Variable Name:** surface_solar_radiation_downwards (ssrd), direct_solar_radiation (fdir).
- **D. Spatial Resolution:** 0.1^\circ \times 0.1^\circ (\sim 9\text{ km} spatial grid).
- **E. Temporal Resolution:** 1-hour hourly accumulation steps.
- **F. Coverage at Nuolja:** 100\% geographic coverage.
- **G. Historical 2024 Availability:** Fully archived on Copernicus Climate Data Store (CDS).
- **H. Archive Access Method:** CDS API retrieval.
- **I. Forecast vs. Analysis/Observation: Global Climate Reanalysis.**
- **J. Forecast Initialization Time:** Non-forecast product; 4D-Var data assimilation cycle.
- **K. Causally Available at t_0: STRICTLY NO.** ERA5 and ERA5-Land are 4D-Var reanalyses that assimilate global surface, radiosonde, and satellite observations across the entire 2024 year. ERA5 fields for t > t_0 incorporate future observations recorded after forecast origin t_0.
- **L. Dissemination Latency:** Released 5 days to 3 months post-observation.
- **M. Already Tested in POLARIS:** No.
- **N. Expected Information Gain:** Unusable for operational benchmarking. Testing ERA5 as a future covariate represents a catastrophic violation of forecasting causality rules.
- **O. Storage/Download Burden:** N/A.
- **P. Technical Extraction Difficulty:** Low.
- **Q. Leakage Risk: CATASTROPHIC.** Total invalidation of benchmark integrity if used as future input.

3. Comparative Master Feasibility Matrix

The candidate data sources were cross-referenced across seventeen technical parameters to establish their archival reality, causal eligibility, and expected utility for breaking the 28\% \text{ nMAE} barrier.

#	Exact Dataset Name	Official Provider	Target Variable	Spatial Res.	Temp. Res.	Nuolja Coverage	2024 Avail.	Access Method	Type (Fcst / Obs)	Causal at t_0?	Dissem. Latency	Tested in POLARIS?	Expected Info Gain	Storage Burden	Tech. Difficulty	Leakage Risk
1	MEPS 3D Vertical (meps_det_vc)	MET Norway	q_l, q_i (Model Lvls 1-65)	2.5 km	1 hour	100 %	Yes	THREDDS NCSS	Fcst	**Yes** (t_{\text{ref}} < t_0)	~3 hours	No	High	450 MB	High	Zero
2	MEPS Surface (meps_det_sfc)	MET Norway	cloud_base/top_altitude	2.5 km	1 hour	100 %	Yes	THREDDS OPENDAP	Fcst	**Yes** (t_{\text{ref}} < t_0)	~3 hours	No	Medium	< 2 MB	Low	Zero
3	MEPS Pressure (meps_det_pl)	MET Norway	U, V (700 hPa Winds)	2.5 km	1 hour	100 %	Yes	THREDDS OPENDAP	Fcst	**Yes** (t_{\text{ref}} < t_0)	~3 hours	Partial	High	15 MB	Low	Zero
4	MET Nordic Grid	MET Norway / SMHI	ssrd (Surface SW)	1.0 km	1 hour	100 %	Yes	SMHI Open Data	Analysis / Fcst	**Partial** (t \le t_0 only)	+45 min	No	Medium	15 GB	Medium	**High** (if t > t_0)
5	NWC SAF PPS Cloud	EUMETSAT / MET Norway	cpp_cot, cpp_cwp	~1 km	Irregular	Intermittent	Partial	THREDDS Swath	Obs	**Conditional** (t_{\text{sat}} \le t_0)	+45 min	No	High (t_0)	120 GB	Very High	Medium
6	NWC SAF HRW Vect	EUMETSAT	amv_speed, amv_	Irregular	Irregular	Sparse	Partial	EUMETSAT Store	Obs	**Conditional** (t_{\t	+60 min	No	High	25 GB	Very High	Low

#	Exact Dataset Name	Official Provider	Target Variable	Spatial Res.	Temp. Res.	Nuolja Coverage	2024 Avail.	Access Method	Type (Fcst / Obs)	Causal at t_0?	Dissem. Latency	Tested in POLARIS?	Expected Info Gain	Storage Burden	Tech. Difficulty	Leakage Risk
	or		dir							ext{obs}} \le t_0)						
7	CAMS Global Composition	ECMWF / CAMS	aod550, duaod550	0.4° (~40 km)	3 hours	100 %	Yes	CAMS ADS API	Fcst	**Yes** (D-1\ text{ 12Z})	~10 hours	No	Low-Med	< 10 MB	Low	Medium
8	CAMS-RAD McClear	ECMWF / Transvalor	ghi_clear, bhi_clear	Point	1-60 min	100 %	Yes	CAMS RAD API	Obs Model	**NO** for H > 0	+1-2 days	No	Zero (Fcst)	< 5 MB	Low	**High**
9	ERA5-Land Reanalysis	ECMWF / C3S	ssrd, fdir	0.1° (~9km)	1 hour	100 %	Yes	Copernicus CDS	Reanalysis	**STRICT NO**	+5 days	No	Zero (Fcst)	N/A	Low	**Catastrophic**

4. Structural Distinctions in Meteorological Data Cataloging

To ensure scientific rigor during data ingestion, candidate variables must be classified into strict hierarchical accessibility tiers. Inferring variable presence from standard name registries often leads to extraction failures when variables do not exist in specific archived runs.
The table below delineates the structural reality of candidate information sources across five critical operational gates:

1. **CF Standard Existence:** Variable standard name defined in Climate and Forecast metadata conventions.
2. **MEPS Native Existence:** Variable output by the HARMONIE-AROME model physics scheme.
3. **2024 Archive Reality:** Variable verified present in the 2024 MET Norway THREDDS repository for Nuolja.
4. **t_0 Causal Usability:** Variable reference time precedes origin time (t_{\text{ref}} < t_0) with documented dissemination.
5. **Benchmark Coverage:** Variable provides complete coverage across all 347 benchmark origins without temporal gaps.

Proposed Information Source	CF Standard Name Defined?	Exists in MEPS Scheme?	Verified in 2024 Nuolja Archive?	Causally Usable at t_0?	Complete Benchmark Coverage?	Final Operational Classification
Model-Level Cloud Water (q_l, q_i)	Yes (mass_fraction_of_cloud_liquid_water_in_air)	Yes (HARMONIE 65-level output)	**YES** (meps_det_vc_2024*.ncml)	**YES** (D-1\text{ 18:00 UTC})	**YES** (347 / 347 origins)	**CAUSALLY ELIGIBLE PREDICTOR**
Diagnostic Cloud Base Altitude	Yes (cloud_base_altitude)	Yes (HARMONIE diagnostic)	**YES** (meps_det_sfc_2024*.ncml)	**YES** (D-1\text{ 18:00 UTC})	**YES** (347 / 347 origins)	**CAUSALLY ELIGIBLE PREDICTOR**
700 hPa Steering Winds (U, V)	Yes (x_wind, y_wind)	Yes (MEPS 3D pressure output)	**YES** (meps_det_pl_2024*.ncml)	**YES** (D-1\text{ 18:00 UTC})	**YES** (347 / 347 origins)	**CAUSALLY ELIGIBLE PREDICTOR**
MEPS Diffuse Shortwave Flux	Yes (surface_diffuse_downwelling_shortwave_flux_in_air)	No (Not written to disk)	**NO** (Forensic audit proved absent)	N/A	N/A	**REJECTED (Archive Deficit)**
MET Nordic Analysis (t > t_0)	Yes (surface_downwelling_shortwave_flux_in_air)	N/A (Spatial Analysis Grid)	Yes (SMHI Open Data Archive)	**NO** (Contains future observations)	Yes	**REJECTED (Future Leakage)**
NWC SAF PPS Cloud Optical Depth	Yes (cloud_optical_depth)	N/A (Satellite Swath Product)	Partial (AVHRR heavy; 2 VIIRS rows)	Conditional (t_{\text{sat}} < t_0)	**NO** (334 / 347 origins; gaps)	**REJECTED (Coverage Gap)**
CAMS 00:00 UTC Run AOD	Yes (atmosphere_extinction_optical_thickness_due_to_ambient_aerosol)	N/A (Global Chemistry NWP)	Yes (Copernicus ADS Archive)	**NO** (Disseminated at 10:00 UTC)	Yes	**REJECTED (Dissemination Latency)**
ERA5-Land Downwelling Shortwave	Yes (surface_solar_radiation_downwards)	N/A (ECMWF Climate Reanalysis)	Yes (Copernicus CDS Archive)	**STRICT NO** (4D-Var Reanalysis)	Yes	**REJECTED (Catastrophic Leakage)**

5. Priority Rankings and Recommended Feasibility Plan

5.1 Top 10 Actually Feasible New Information Sources

The following ten information sources represent verified, causally clean data candidates capable of being ingested into the POLARIS benchmarking framework:

1. **MEPS Native Model-Level Cloud Liquid/Ice Mass Fractions (q_l, q_i on Hybrid Levels 1–65):** Boundary-layer cloud water resolution directly above Nuolja's topography (675\text{ m a.s.l.}) without pressure-level interpolation distortion.
2. **MEPS Boundary-Layer / Tropospheric Steering Winds (U, V at 700\text{ hPa} / Hybrid Levels 30–45):** Free-atmosphere wind direction and magnitude at cloud-layer height to project directional cloud movement.
3. **MEPS Diagnostic Cloud Base and Cloud Top Altitudes (cloud_base_altitude, cloud_top_altitude):** Yields physical cloud layer thickness (\Delta h) and sub-cloud clearance depth.
4. **CAMS Global Atmospheric Composition Forecasts (D-1\text{ 12:00 UTC} AOD Run):** Incorporates real-time aerosol optical depth variations (aod550, duaod550) for clear-sky attenuation.
5. **MET Nordic 1.0 km Pre-Forecast Ground Context (t \le t_0 Surface Solar/Temperature Analysis):** Captures 1.0 km high-resolution surface weather states preceding forecast origin.
6. **EUMETSAT NWC SAF PPS Cloud Optical Thickness (cpp_cot at t_{\text{sat}} \le t_0):** Ingests satellite-derived optical depth immediately prior to forecast origin.
7. **EUMETSAT NWC SAF PPS Liquid Water Path (cpp_cwp at t_{\text{sat}} \le t_0):** Ingests satellite-derived liquid water path prior to origin.
8. **MEPS Spatial Neighborhood 3D Cloud Water Profiles (3 \times 3 Grid Neighborhood on Hybrid Levels):** Samples upstream cloud water content in a 2.5 km grid ring around Nuolja.
9. **EUMETSAT NWC SAF High-Resolution Winds (HRW Atmospheric Motion Vectors at t \le t_0):** Observes satellite-tracked cloud displacement vectors in the Scandinavian sector.
10. **MEPS Model-Level Relative Humidity and Specific Humidity (q, \text{RH} on Lowest 15 Hybrid Levels):** Resolves sub-cloud condensation state and boundary-layer moisture flux.

5.2 Top 5 Highest-Value Sources

Selecting from the feasible sources, five candidates offer the strongest physical mechanisms for error reduction:

Ranking	Information Source	Target Variable	Primary Physical Contribution	Technical Feasibility
1	MEPS Model-Level Cloud Profiles	q_l, q_i (Levels 1–20)	Resolves low-level mountain stratus directly above Nuolja surface (675\text{ m}).	High (THREDDS NCSS)
2	MEPS 700 hPa Steering Winds	U_{700}, V_{700}	Projects upwind cloud displacement	High (THREDDS OPENDAP)

Ranking	Information Source	Target Variable	Primary Physical Contribution	Technical Feasibility
			relative to Lake Torneträsk / Mt. Nuolja ridge.	
3	MEPS Diagnostic Cloud Geometry	Cloud Base / Top Altitudes	Informs GBDT of physical cloud column thickness (\Delta h) and light scattering path.	Very High (Surface file)
4	NWC SAF PPS Satellite COT	cpp_cot (t_{\text{sat}} \le t_0)	Observes pre-forecast satellite cloud optical depth prior to model execution.	Medium (Intermittent overpasses)
5	CAMS D-1\text{ 12Z} Global AOD	aod550, duaod550	Modulates clear-sky baseline during Siberian/Canadian wildfire smoke transport events.	High (Copernicus ADS API)

5.3 Top 3 Sources with the Best Probability of Materially Reducing n\text{MAE}

1. **MEPS Native Model-Level Cloud Hydrometeor Profiles (q_l, q_i on Hybrid Levels 1–20):** High probability of reducing error because standard pressure levels (1000, 925, 850 hPa) severely distort cloud profiles over Nuolja's 675\text{ m} mountain surface. Ingesting native hybrid levels directly captures boundary-layer cloud water without interpolation artifacts.
2. **MEPS Cloud-Level Steering Wind Vectors (U, V at 700\text{ hPa} / Hybrid Level 35):** Enables the decision tree model to weigh spatial cloud water content based on whether winds blow from Lake Torneträsk (east) or Mt. Nuolja ridge (west).
3. **MEPS Diagnostic Cloud Geometry (\Delta h = \text{Cloud Top} - \text{Cloud Base}):** Directly informs the model of physical cloud thickness, addressing cases where total column water is identical but concentrated in thin dense stratus versus thick diffuse altostratus.

5.4 Recommended First Feasibility Audit

Audit Specification: MEPS Native Model-Level Archive Existence and t_0 Causality Audit for 2024 Nuolja Origins.
The objective of this audit is to verify that native model-level 3D files exist on the MET Norway THREDDS server for all 347 benchmark origins in 2024, confirm dissemination latency, and test dynamic vertical coordinate reconstruction.

Audit Step	Operational Action	Verification Target	Success Criteria
Step 1	Query MET Norway	https://thredds.met.no/t	Confirm presence of

Audit Step	Operational Action	Verification Target	Success Criteria
	THREDDS Repository	hredds/catalog/meps25eps/ catalog	meps_det_vc_2024*.ncml for all 347 origins.
Step 2	Audit Reference Timestamp t_{\text{ref}}	NetCDF global attributes (forecast_reference_time)	Verify t_{\text{ref}} = D-1\text{ 18:00 UTC} < t_0 across 100% of origins.
Step 3	Point-Extraction Stress Test	Extract q_l, q_i, p_{\text{sfc}} at grid point y=760, x=486	Ensure zero missing values, zero NaNs, and complete 65-level records.
Step 4	Hybrid Level Pressure Evaluation	p(k) = a(k) + b(k) \cdot p_{\text{sfc}} for levels k=1 \dots 65	Confirm dynamic pressure calculation evaluates strictly above surface pressure (\sim 930\text{ hPa}).

5.5 Recommended First Forecasting Experiment

Experiment Specification: POLARIS Stage 8.1 — Native Model-Level Hydrometeor and Steering Wind Feature Integration.
The objective of Stage 8.1 is to test whether incorporating 6 pre-specified model-level cloud and steering wind features into the frozen Stage 7.5 GBDT architecture achieves a statistically significant reduction in n\text{MAE} across 5-fold forward-chaining cross-validation.

Experimental Parameter	Locked Experimental Specification
Control Baseline (E_0)	Frozen Stage 7.5 Champion (35 features; EVEN-CV MAE = 37.835137\text{ W/m}^2, n\text{MAE} = 38.5174\%).
Candidate Feature Set (E_1)	Augment E_0 with 6 candidate features: meps_ml_lwp_boundary (integral q_l levels 1–20), meps_ml_iwp_boundary (integral q_i levels 1–20), meps_ml_q_liquid_peak (max q_l levels 1–20), meps_ml_cloud_base_alt, meps_u_wind_700, meps_v_wind_700.
Model Algorithm	HistGradientBoostingRegressor(loss="absolute_error", max_depth=4, min_samples_leaf=30, l2_regularization=10.0, random_state=42).
Development Split	174 EVEN development origins evaluated across 5 forward-chaining folds.
Acceptance Gate 1 (MAE)	EVEN-CV Daytime MAE \le 37.635137\text{ W/m}^2 (\ge 0.20\text{ W/m}^2 improvement over Stage 7.5).
Acceptance Gate 2 (Folds)	Daytime MAE improved in \ge 4 of 5 forward-chaining folds.
Acceptance Gate 3 (Degradation)	Maximum single-fold degradation \le +0.15\text{ W/m}^2.
Acceptance Gate 4 (nMAE)	EVEN-CV n\text{MAE} \le 38.5174\%.

Experimental Parameter	Locked Experimental Specification
Acceptance Gate 5 (RMSE)	EVEN-CV RMSE \le 60.307218\text{ W/m}^2.
ODD Governance Lock	ODD test set (173 origins) remains **100% UNTOUCHED & LOCKED** until EVEN gates are passed.

6. Data-Source Risks, Dead Ends, and Theoretical Horizons

6.1 Data-Source Risks and Structural Dead Ends

1. **CAMS Aerosol Optical Depth in Arctic Regimes (Likely Dead End):** Background aerosol optical depth at Nuolja is extremely low (\text{AOD}_{550} \sim 0.04). Variations in AOD under non-wildfire conditions modulate shortwave surface flux by less than 2\text{ W/m}^2. Investing technical effort into CAMS API integration is highly unlikely to produce statistically significant reductions in daytime MAE when cloud-driven errors exceed 60\text{ W/m}^2.
2. **Geostationary Satellite Imagery at 68.36^\circ\text{N} (Structural Dead End):** Geostationary satellites (Meteosat SEVIRI / FCI) are positioned over the equator (0^\circ\text{N}). At Nuolja (68.36^\circ\text{N}), the satellite viewing zenith angle (VZA) exceeds 76^\circ. This causes severe parallax errors (> 15\text{ km} spatial distortion), extreme pixel elongation, and atmospheric path attenuation. Geostationary optical depth products are physically unusable at this latitude.
3. **MET Nordic Analysis Fields for Future Horizons (Causal Dead End):** Attempting to ingest MET Nordic 1.0 km spatial analysis fields for forecast lead times H1 \dots H24 represents a catastrophic data leakage risk. MET Nordic analyses incorporate real-time station observations and radar data post-facto. Using them as future covariates invalidates the benchmark.
4. **Coarse Reanalysis Products (ERA5 / ERA5-Land) (Causal Dead End):** Reanalysis products assimilate global surface and satellite observations recorded across the target period. ERA5 fields at t > t_0 contain future information and are strictly ineligible for operational forecasting.
5. **Raw 3D Grid Spatial Extraction Bandwidth Bloat (Technical Risk):** Downloading full 3D spatial grids (65\text{ levels} \times 100 \times 100\text{ grid points}) for all MEPS forecast steps requires terabytes of storage and causes severe network throttling on THREDDS servers. Spatial extractions must be strictly constrained to targeted point profiles and small 3 \times 3 grid neighborhoods.

6.2 Theoretical Framework and Data Requirements for Sub-15% n\text{MAE}

Achieving a daytime n\text{MAE} < 15\% (< 19.3\text{ W/m}^2 \text{ MAE} at Nuolja) represents a fundamental shift from regional NWP parameterization to real-time microscale cloud advection tracking. Regional NWP models (including MEPS 2.5 km) suffer from spatial timing errors: an NWP model may correctly forecast a cloud deck, but arrive 45 minutes early or late, resulting in temporary irradiance errors exceeding 300\text{ W/m}^2.

Forecast Lead Time	Dominant Error Mechanism	Required Information Source	Required Hardware / Data Infrastructure
H1 \dots H3 (Nowcasting)	Cloud timing and optical depth misplacement	High-frequency ground cloud vector tracking	All-Sky Imagers (ASI) & Doppler Cloud Radar
H3 \dots H6 (Short-Range)	Sub-grid mesoscale advection tracking	High-resolution satellite optical thickness	Rapid-scan polar swath satellite retrievals
H6 \dots H24 (Day-Ahead)	Microscale boundary layer cloud formation	Cloud-resolving microscale NWP / LES	Sub-kilometer Large-Eddy Simulations (LES)

To break 15\% \text{ nMAE}, the forecasting system requires:

- **High-Frequency Ground-Based Sky Imaging:** Ground-based All-Sky Imagers (ASI) at Nuolja capturing hemispheric cloud fraction, cloud optical depth, and local cloud velocity vectors at 1-minute intervals. ASI features eliminate NWP cloud timing errors for H1 \dots H3 horizons.
- **Local Ceilometer / Lidar Profiling:** Continuous vertical profiling of cloud base height and optical thickness directly upwind of the solar disk.
- **Sub-Kilometer Microscale NWP / LES:** Large-Eddy Simulations (LES) or cloud-resolving micro-NWP models running at \le 500\text{ m} spatial resolution over the Abisko alpine terrain to resolve valley-scale fog and orographic cloud formation.

6.3 Physical Requirements to Approach 10% n\text{MAE}

A daytime n\text{MAE} approaching 10\% (\sim 12.9\text{ W/m}^2 \text{ MAE}) approaches the intrinsic physical limit of predictability for surface solar irradiance in mountainous Arctic terrain. At this threshold, atmospheric radiative transfer is governed by three-dimensional photon scattering effects (e.g., cloud-edge refraction, mountain snow-albedo multi-reflection) that violate standard 1D plane-parallel radiative transfer assumptions.
Approaching 10\% \text{ nMAE} requires three mandatory technological pillars:

1. **Direct Solar Disk Tracking and Circumsolar Radiometry:** Real-time optical monitoring of the solar narrow-angle circumsolar region (5.8^\circ field of view) to detect thin ice-crystal cirrus and sub-visible aerosol extinction.
2. **Real-Time 3D Cloud Tomography:** Multi-camera ground networks or stereoscopic dual-satellite polar overpasses capable of reconstructing the full 3D geometrical volume and optical density (\tau_x, \tau_y, \tau_z) of approaching cloud fields.
3. **Dynamic 3D Monte Carlo Radiative Transfer Coupling:** Replacing standard GBDT/NWP 1D clear-sky/cloud attenuation models with real-time 3D Monte Carlo photon transport simulations incorporating 30m digital elevation topography, anisotropic snow surface bidirectional reflectance distribution functions (BRDF), and 3D cloud volume structures.

7. Operational Conclusions and Next Steps

The POLARIS Stage 7.5 champion represents the maximum performance extractable from point-based ground context, 1D pressure-level hydrometeor integrals, native surface NWP diagnostics, and static topographic horizon geometry (28.3983\% \text{ ODD nMAE}, 36.613571\text{ W/m}^2 \text{ MAE}). Incremental adjustments to surface features or ratio target

reformulations have proven incapable of producing statistically significant out-of-sample improvements.

To break the 28\% \text{ nMAE} barrier, POLARIS must transition to extracting 3D native model-level cloud profiles (q_l, q_i on hybrid levels 1–65) and mid-troposphere steering winds (U_{700}, V_{700}) from the MET Norway MEPS vertical archive. This approach directly targets the primary physical bottleneck—boundary-layer cloud mass resolution and directional advection—while maintaining 100\% historical coverage and causal eligibility across all 347 benchmark origins in 2024.